Computational reproducibility in nonlinear shell buckling: a crowdsourced cross-platform Abaqus benchmark

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Abstract

Verification of numerical models is central to safety-critical finite element-based structural design, for which the new EN 1993-1-14 now provides explicit provisions. However, an interesting issue concerns whether these provisions should offer cross-platform reproducibility guarantees. This short communication reports on a crowdsourced benchmark performed in 2026 that demonstrates macroscopically different non-linear equilibrium path families arising from serial execution of identical Abaqus input files on different machines and installations. The benchmark is a cylindrical shell under uniform meridional compression, with a control case (plastic collapse) that was expected to be insensitive to the execution platform and an interesting case (elastic buckling) which was expected to be highly platform sensitive. Three dominant response families were found in the dataset, aligned with CPU vendor and Abaqus release, of which only one follows the expected fundamental path for the highlighted nonlinear elastic buckling case over the computed range. The variability when quantified using pairwise Fréchet distances appears genuine, but the crowdsourced dataset is too small and self-selected to support a causal attribution. The findings raise questions about the reproducibility of nonlinear structural predictions near unstable equilibrium paths, and also illustrate why agreement between computing platforms does not itself establish correctness. We release the benchmark and anonymised data in the hope of starting a discussion and facilitating a larger controlled cross-platform and cross-solver experiment covering additional benchmarks at a future time.

Keywords: computational reproducibility, nonlinear finite element analysis, computational shell buckling

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1. Introduction

The use of finite element analysis in steel structural design places considerable responsibility on verification of the computed results. Through its Clauses 4.2.7(4) and 4.2.8(3), EN 1993-1-6:2025 [5] on the strength and stability of metal shells explicitly refers both geometrically nonlinear elastic analysis (GNA) and materially nonlinear analysis (MNA) respectively to the verification and validation criteria of EN 1993-1-14:2025 [6]. These include the usual good-practice checks of discretisation and load-increment sensitivity, together with engineering assessment of the resulting deformations and load-displacement paths (see for example Clauses 7.2(1), 7.2(3) and 7.2(9) therein). Benchmark comparisons also provide a means of assessing the numerical model and its solution settings (Clauses 7.3(1)–(2)), although for metal shell structures there are very few published fully worked examples suitable for this purpose [8].

A related but apparently underexplored question is whether a given benchmark result is actually reproducible across different computing platforms even when the input files are unchanged and the same software package is used. We explore precisely this reproducibility question in the present short paper, submitted to the special issue 'Advanced FEM-based design of steel structures according to EN 1993-1-14', using a unique crowdsourced nonlinear shell buckling benchmark. We present the findings and data to the community so that it is aware that cross-platform reproducibility currently cannot be guaranteed with commercial finite element software.

2. The benchmark

It all started with a simple task, to execute an Abaqus input file unchanged on a different machine. The task was part of an undergraduate assignment in computational shell buckling at Imperial College London, where students were modelling a cylindrical shell under uniform meridional compression using increasingly sophisticated finite element analyses. An early task was a materially nonlinear but geometrically linear analysis (MNA) under an elastic-perfectly plastic material model where a distributed edge load per unit circumference of magnitude equal to the plastic collapse load $N_y = \sigma_y t$ was applied (σ_y is a yield stress, t is a unit thickness). This was to give an initial linear path followed by an indefinite plateau at a normalised load proportionality factor of unity. A collapse mechanism should form first through plastic hinging in the boundary layers, followed by the spread of membrane yielding through the rest of the shell (Fig. 1, left). A later task was an unperturbed geometrically nonlinear but elastic analysis (GNA) under

an edge load of magnitude equal to the classical elastic critical buckling load $N_{cl} = \sigma_{cl} t \approx 0.605 E t^2/r$ (assuming a Poisson ratio of 0.3, E is a Young's modulus and r is a midsurface radius), intended to trace the fundamental equilibrium path of this system. Less obviously and much less universally appreciated (Section 9.5a in [2]), this response should likewise form a plateau at a load proportionality factor of unity as the bending boundary layers grow without bound and fully elastic sinusoidal deformations penetrate into the shell (Fig. 1, right). No formal geometric imperfections of any kind were introduced. Things got interesting when students obtained neither expected response consistently, but instead obtained mostly non-physical results that varied between machines.

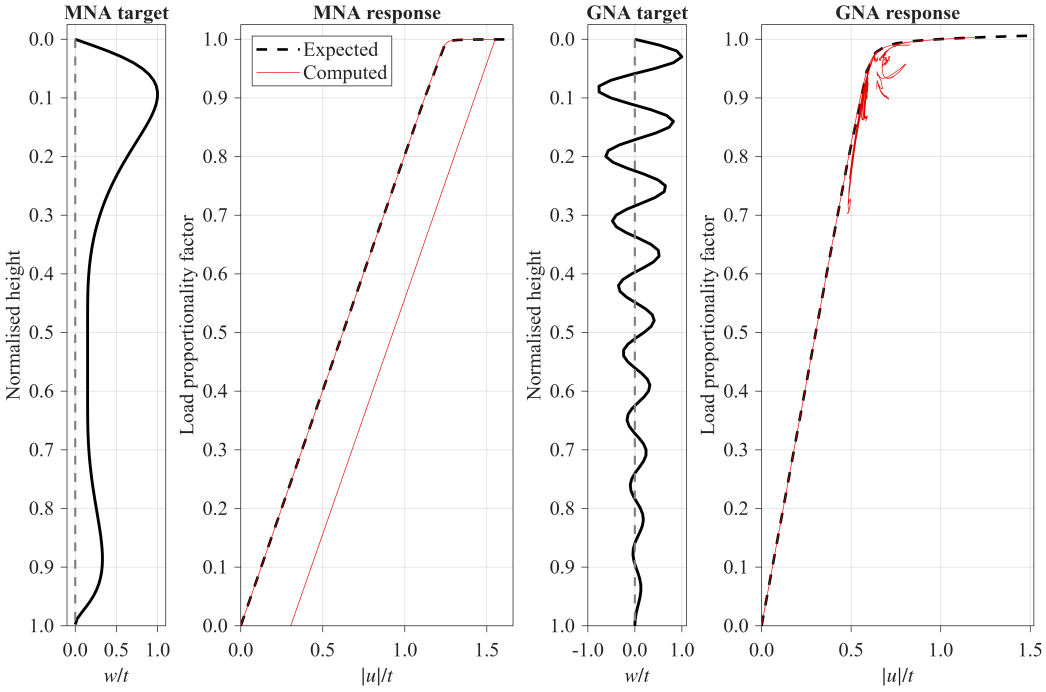


Figure 1: Target deformation profiles and corresponding response paths for the materially nonlinear analysis (MNA) and geometrically nonlinear analysis (GNA) benchmark cases. The 'target' panels show the normalised radial deformation w/t over the normalised height, 'response' panels show the load proportionality factor against the normalised loaded-edge meridional shortening $|u|/t$. Here t is a unit thickness.

The idea of a systematic reproducibility benchmark crystallised from this experience, and a public call to crowdsource a larger dataset was made. The benchmark consisted of four small, near-identical Abaqus input files. One was an MNA 'control' which was not intended to be interesting (i.e. it was expected to give the same macroscopic response across all execution

platforms), and three GNA files differing only by the initial and maximum arc-length increment sizes. Participants were asked to run all files to completion, serially and without modification, then execute a Python script to extract loading histories, and document the hardware (CPU, memory) and software (Abaqus build, OS) environments in a metadata questionnaire.

3. Results

A total of 33 submissions were received whose distinct load histories are illustrated in Fig. 1. All MNAs exhibit the correct initial linear response, briefly follow a plateau approaching the expected load level but then erroneously reverse the direction of loading. This reversal is a manifestation of the 'doubling back' pathology [3] known to occur in arc-length nonlinear solvers, but at least it was obtained consistently by all submissions. The GNAs, however, despite following the correct initial near-linear response eventually diverged into multiple macroscopically different late-stage paths as highlighted in Fig. 2, *most* of them incorrectly departing from the expected plateauing fundamental path. The fundamental GNA equilibrium becomes energetically unstable after the first bifurcation point, which for the benchmark's boundary conditions should occur at a normalised load level of approximately 0.9 (Figure 2.44 in [9] for $Z \approx 10^3$, C1-S3). Absent an explicit perturbation of the initial geometry (the standard technique for triggering bifurcations by transforming them into very sharp limit points), or some other source of perturbation, the solver should be expected to be able to trace this unstable but smooth fundamental path correctly.

Deterministic SHA256 fingerprints identified 12, 11, 14 and 11 unique histories for MNA, GNA1, GNA2 and GNA3, respectively. Pairwise multivariate Fréchet distances [4] (which preserve the ordering of points along each history) were computed between these unique histories and are shown as colour-coded matrices in Fig. 3 for the MNA and GNA2 cases only. MNA histories, intended to be an exactly reproducible control, unexpectedly contained microscopic differences. That these still produce the same (albeit erroneous) macroscopic plastic collapse response is consistent with MNAs being perturbation insensitive and thus appropriate for use as a macroscopic 'control', for this system at least. By contrast, the GNA responses exhibit severe variability, with microscopic differences cascading to give macroscopically radically different histories, apparently from platform-specific numerical causes. This is consistent with the known sensitivity of thin cylinders under meridional compression to imperfections (Sections 4.6 & 4.7 in [9]), and generally with the sensitivity of unstable nonlinear systems to perturbation.

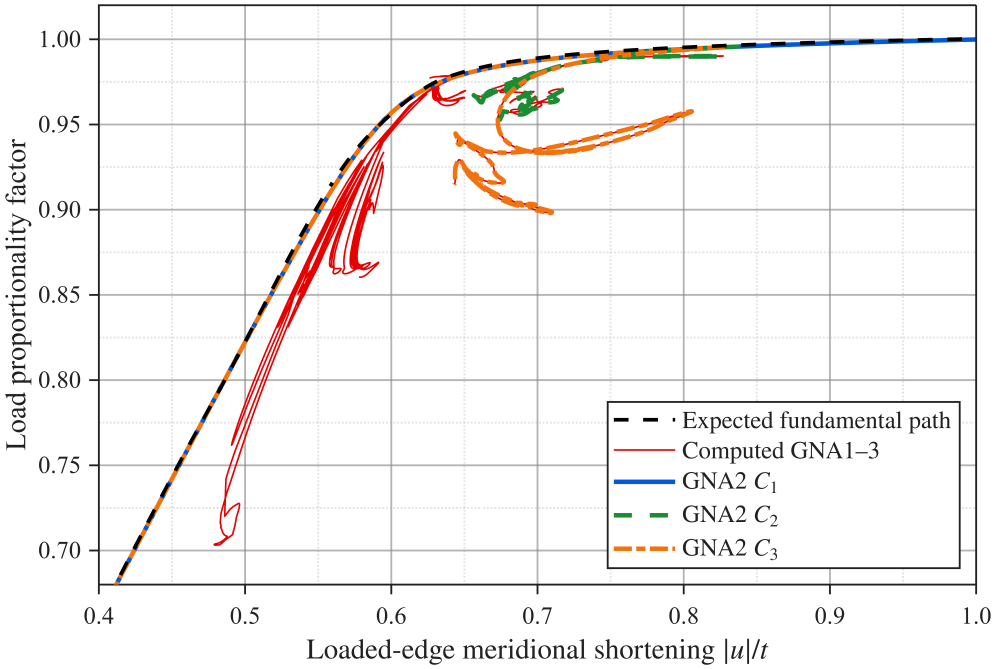


Figure 2: Enlarged view of the GNA response paths near the expected plateau. Three dominant GNA2 response families C_1-C_3 are highlighted with colours matching the dendrogram in Fig. 4. The dashed black curve shows the expected fundamental equilibrium path; only C_1 follows it over the computed range.

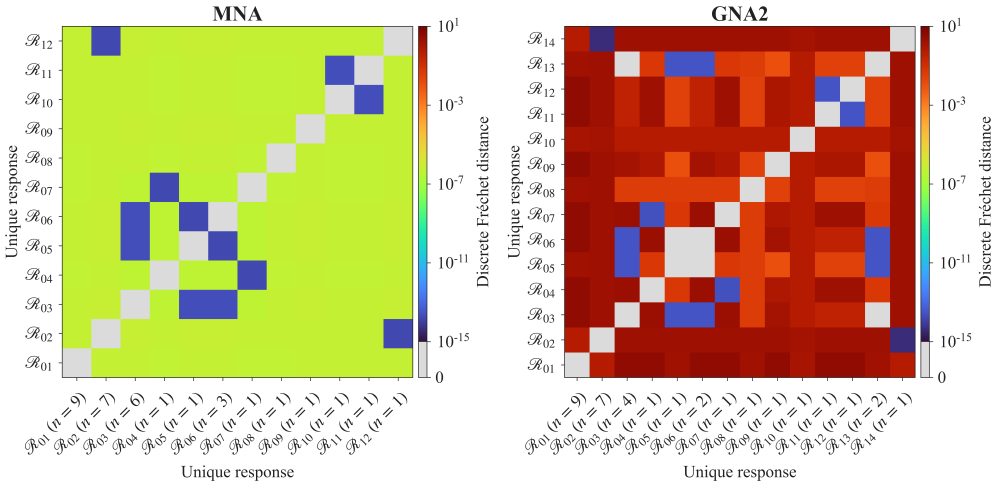


Figure 3: Pairwise discrete Fréchet distances between the unique MNA and GNA2 histories represented by the standardised load proportionality factor, loaded-edge meridional shortening and maximum force residual (missing values are omitted from pointwise comparisons). Each coordinate is scaled by its standard deviation across the concatenated unique histories for that analysis. Colours show distance on a common logarithmic scale, grey denoting zero. n is the number of submissions for each unique response $\mathcal{R}_i$.

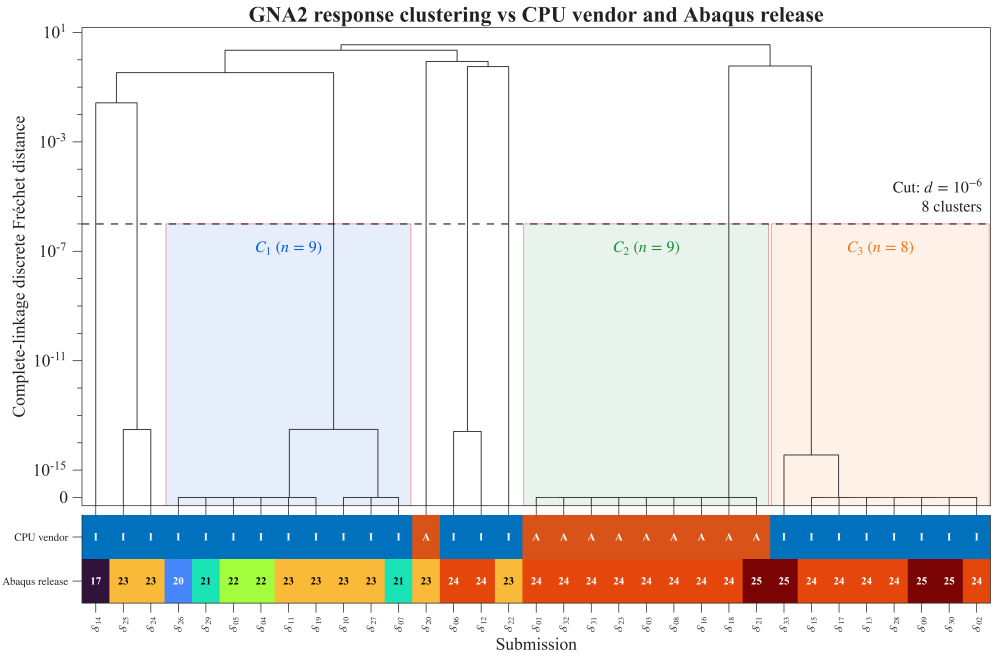


Figure 4: Dendrogram of GNA2 response histories based on pairwise multivariate Fréchet distances with complete linkage. Submissions $\mathcal{S}_i$ are labelled by CPU vendor and Abaqus release. A reasonable cut at $d = 10^{-6}$ highlights three dominant clusters, two doubletons and three singletons. Responses of clusters C_1, C_2 and C_3 are highlighted in Fig. 2.

Recurring GNA paths indicate structured, partly reproducible variability. Complete-linkage clustering, which favours internally compact clusters [1], performed on the basis of the pairwise Fréchet distances between response histories (i.e. *not* on the basis of additional questionnaire metadata) suggests three dominant families of submissions in the dataset. These are Intel CPUs with older Abaqus versions 2020-23 (C_1, nine submissions), AMD CPUs with newer Abaqus versions 2024-25 (C_2, nine submissions) and Intel CPUs with newer Abaqus versions 2024-25 (C_3, eight submissions). The remaining seven submissions are, in this dataset, distributed across three singleton and two doubleton clusters. These GNA2 clusters are illustrated in Fig. 4 and it may be noted that they remain unchanged across several orders of magnitude of Fréchet distance threshold, indicating a persistent separation between response families. Of these, only C_1 for GNA2 actually follows the expected response (Fig. 2). The memberships of the AMD and newer-Intel dominant clusters remain unchanged across MNA, GNA1 and GNA3, while the older-Intel dominant cluster varies only slightly. This recurrence of clusters across the four analyses is consistent with a platform-dependent hypothesis, but the small, self-selected dataset does not separate the effects of hardware, software

release and other execution-environment differences sufficiently to support a causal attribution. It is acknowledged that at least part of the microscopic variability may still be attributable to export-formatting effects.

Lastly, the variability in the peak load proportionality factor across the four individual analyses is illustrated in Fig. 5. It is reminded that the correct result representing the 'peak' of either plateauing fundamental equilibrium path is a load proportionality factor of unity. The MNA control is essentially spot on this value with no dispersion, despite the later 'doubling back' numerical failure. The GNA results, while close, are *not* actually at unity and in places exhibit non-negligible dispersions arising solely from cross-platform execution of an identical input file. Moreover, both the median and dispersions are functions of the load increment size, the discriminator between the three GNA analyses which assume 0.01, 0.03 and 0.05 initial and maximum arc-length increments for GNA1, GNA2 and GNA3 respectively. The varying load increment size is thus an independent source of additional variability, and evidently a reduced load increment does *not* necessarily lead to improved agreement with a reference result or reduced cross-platform dispersion (an effect that has been observed elsewhere e.g. [7]).

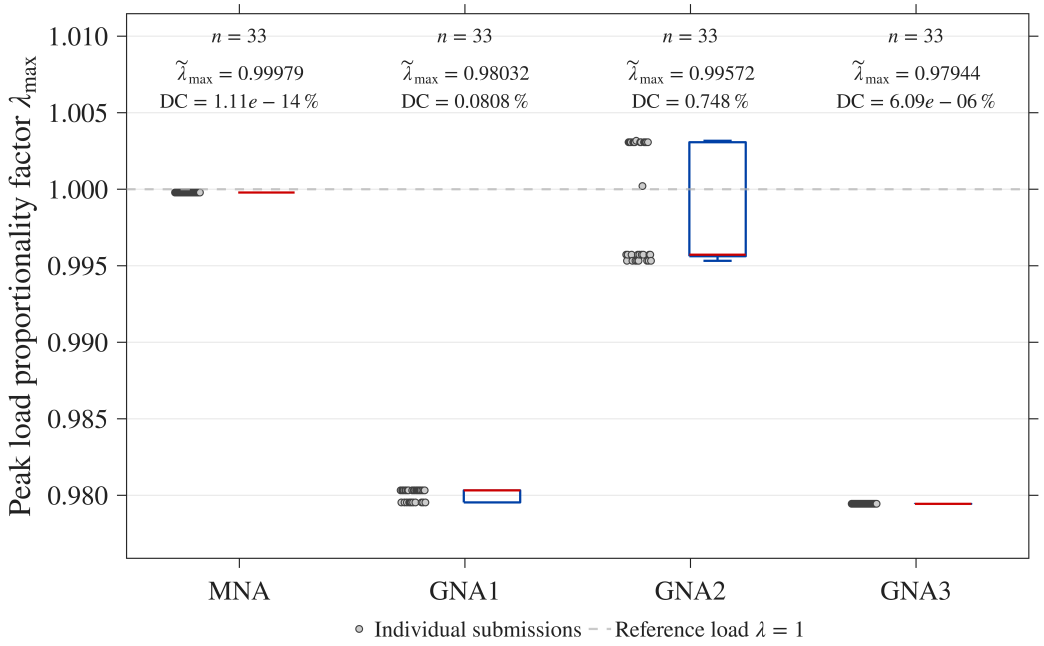


Figure 5: Box-and-whisker plots of the recorded peak load proportionality factors λ_{max} for MNA and GNA1-3. Boxes span the first to third quartiles, red lines denote medians, and whiskers extend to the minimum and maximum. The dispersion coefficient is DC = $100(Q_3 - Q_1)/\tilde{\lambda}_{max}$, expressed as a percentage.

4. Conclusions

This short communication reports on a crowdsourced benchmark performed in 2026 that illustrates how unchanged Abaqus input files produced recurring and macroscopically distinct nonlinear responses correlated with the computational platform. In short:

- Submissions of a 'control' materially nonlinear analysis (MNA) successfully produced a consistent macroscopic response across all submissions despite microscopic differences in the load histories. However, this agreement is clearly no guarantee of correctness, as all MNA submissions erroneously reversed the direction of loading after reaching the expected plateau as part of an unrelated 'doubling back' pathology with the 'Riks' arc-length solver implementation of Abaqus.

- Submissions of a geometrically nonlinear analysis (GNA) exhibited severe and apparently structured platform-dependent variability, of which only a small subset was actually correct.

- The recorded peak loads were much more consistent than the complete equilibrium paths, so a focus on benchmarking a single scalar result can mask substantial differences in nonlinear responses. In the present GNA data, both the peak values and their dispersions were found to vary with the prescribed arc-length increment, and a reduction in this does not necessarily lead to improved agreement with the expected fundamental equilibrium path.

- The system under investigation, a thin cylindrical shell under uniform meridional compression, is very well understood and both MNA and GNA benchmarks have known analytical solutions. This made identifying incorrect responses straightforward. For less well-understood nonlinear structural systems, such solver- or platform-dependent pathologies may be much harder to identify, let alone resolve, and the present findings raise questions about just how reproducible nonlinear structural predictions are in general.

The authors call for a larger controlled cross-platform and cross-solver reproducibility study on more safety-critical structural systems of industrial significance to raise awareness and accurately identify correlations, causal mechanisms and countermeasures. A database of 'sensitive' benchmarks such as this one would be a particularly valuable resource for the community.

Declarations

CRediT authorship statement. Yuhan Wei: Formal analysis, Writing - original draft. Molly Balassa: Conceptualization, Formal analysis, Writing - original draft. Adam J. Sadowski: Conceptualization, Formal analysis, Writing - original draft.
Data and code availability. Anonymised data, benchmark materials and processing code are available at `https://github.com/SadowskiAJ/RRX_public.git`. There is significantly more in the data than has been reported on here.
Declaration of generative AI use. OpenAI Codex was used to assist with development of the analysis code. All resulting code was reviewed by the authors.
Competing interests. The authors declare no competing interests.
Acknowledgements. The authors gratefully acknowledge the participation of all crowdsourcing contributors. Dr Sadowski acknowledges EPSRC Fellowship funding under EP/Z533622/1, Mx Balassa acknowledges an EPSRC ICASE studentship. Dr Sadowski acknowledges a particularly fruitful discussion with his colleague Prof. Peter Stafford on the results and on the wider context of software-level variability and reproducibility.

References

[1] M.S. Aldenderfer and R.K. Blashfield (1984) "Cluster analysis", Sage College Publishing.

[2] D.O. Brush and B.O. Almroth (1975) "Buckling of bars, plates and shells", McGraw-Hill.

[3] M.A. Crisfield (1981) "A fast incremental/iterative solution procedure that handles 'snap-through"', *Computers & Structures* 13(1-3):55-62. `https://doi.org/10.1016/0045-7949(81)90108-5`.

[4] T. Eiter and H. Mannila (1994) "Computing discrete Fréchet distance", *Technical Report CD-TR 94/64*, TU Vienna. `https://www.kr.tuwien.ac.at/staff/eiter/et-archive/files/cdtr9464.pdf`

[5] EN 1993-1-6 (2025) "Eurocode 3: Design of Steel Structures. Part 1-6: Strength and Stability of Shell Structures" Comité Européen de Normalisation (CEN), Brussels.

[6] EN 1993-1-14 (2025) "Eurocode 3: Design of Steel Structures. Part 1-14: Design assisted by finite element analysis" Comité Européen de Normalisation (CEN), Brussels.

[7] A.J. Sadowski and A. Filippidis (2025) "The capacity curve framework of EN 1993-1-6 (2025): Cylindrical shells under uniform meridional compression vs uniform bending" *Journal of Constructional Steel Research*, 109167. `https://doi.org/10.1016/j.jcsr.2024.109167`

[8] A.J. Sadowski and M. Seidel (2023) "8-MW wind turbine tower computational shell buckling benchmark. Part 2: Detailed reference solution" *Engineering Failure Analysis*, 107133. `https://dx.doi.org/10.1016/j.engfailanal.2023.107133`

[9] N. Yamaki (1984) "Elastic stability of circular cylindrical shells", North-Holland.